Result 1 (basis at a restricted vertex). [Clean topology study, basis at a restricted vertex]
Let K be a type. Let *field* be a field structure on K. Let M be a module over K. Let I be a type. Let n, a and m be natural numbers. Assume $a+m \leq n$ (`bound`). Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let v be a vertex in $\{0, \ldots, m+1-1\}$. Let b be a basis of M indexed by $A(L, a+v)$. Let *vectors* be a function from I to M. Assume (for every $x : A(L, a+v),\ b_x = \mathit{vectors}(x)$) (`realizes`). Then

$$(\text{there exists } c : \mathrm{Basis}_K(M, R(L, v)),\ (\text{for every } y : R(L, v),\ c_y = \mathit{vectors}(y)))\,.$$

LLM-assisted statement and proof (gpt-6, 2026-11-07). See PROVENANCE.md. This is the pointwise basis transport result; assembling an interval basis and proving edge compatibility remain separate obligations.